

Ritvik Singh

ritvikchandarsingh@gmail.com | (916) 260-6616 | linkedin.com/in/singhritvik | github.com/ritviks123 | Folsom, CA

EDUCATION

University of California, Irvine - Irvine, CA

Bachelor of Science in Computer Engineering | GPA: 3.83 / 4.00

Expected May 2028

- **Relevant Coursework:** Advanced C Programming, Introductions to Digital Systems, Digital Logic Design Lab, Organization of Digital Computers, Computer Systems and C Programming, Drones, Boolean Logic and Discrete Structures, Multivariable Calculus, Differential Equations, Linear Algebra

EXPERIENCE

Zotbins - Irvine, CA

August 2026 - Present

Electrical Engineering Intern, Hardware Subteam

- Architecting a solar power and energy storage subsystem driving ZotBin servos off-grid, retiring the wall-AC tether
- Designed and hand-assembled a 2-layer KiCad PCB (battery input, BJT/MOSFET switching, custom footprints)
- Evaluated scoping with 3 EECS faculty, steering the team toward hardware plus formal research deliverables
- Scoping an MPPT study under repeatable partial shading to quantify harvested energy vs. a fixed-duty baseline

IEEE at UC Irvine, Open Project Space - Irvine, CA

September 2025 - June 2026

Member

- Engineered 7 guided builds plus a capstone spanning embedded firmware, analog circuits, and wireless IoT
- Mastered KiCad schematic capture, SMD rework, and oscilloscope debugging, applied across every project below
- Attended weekly lectures preceding each build, covering microcontrollers, sensor interfacing, and PCB layout

PROJECTS

F.R.N.D. - Serverless Emergency Drone Network

Arduino, Raspberry Pi, Python, Llama3

IrvineHacks 2026, Qualcomm Sponsor Track | March 2026

- Deployed a quantized Llama on the Arduino Uno Q weeks post-launch, running inference in-process via yzma
- Engineered the WebSocket relay bridging Uno Q and Raspberry Pi, splitting phone and model clients by URL path
- Delivered a zero-infrastructure emergency mesh with a 4-person team in 3 days, no app or internet required

Autonomous RC Rover with Custom PCB

C++, KiCad, Embedded Systems

IEE Operation Project Space Capstone | April 2026

- Engineered a 2-layer custom PCB consolidating 5 discrete modules onto a single board by routing in KiCad with proper ground planes and impedance control to fit financial and power constraints
- Implemented a PID feedback loop over IR and ultrasonic error, driving dual motors via an L293D H-bridge
- Tuned Kp and Kd empirically against serial error traces, achieving a 90% obstacle-avoidance rate in testing

IoT Weather Station

ESP32, MQTT, C++, I2C

Personal Project | January 2026

- Extended battery runtime 9x (3 days to 4 weeks) by isolating I2C timing faults and scheduling deep-sleep cycles
- Streamed DHT22 temperature and humidity telemetry over MQTT at sub-2-second sensor-to-dashboard latency

SKILLS & INTERESTS

Programming: Python, Java, JavaScript, C, C++, HTML, Verilog

Hardware & Embedded: ESP32, Arduino, Raspberry Pi, Circuit Design, SMD Soldering, Oscilloscope, Multimeter

Design & CAD: KiCad, SolidWorks, Fusion 360

Tools & Protocols: AWS, Git, Linux, MATLAB, MQTT, I2C, SPI, SQL

Certificates: AWS Certified Solution Architect - Associate Level